

Performance Testing

Date	2 July 2026
Team ID	SWTID-2026-2260
Project Name	Smart Lender
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman, Browser (Google Chrome), Flask Development Server
Type of Testing	Load Testing and Performance Testing
Target Module / API	Loan Eligibility Prediction Module (/predict)
Test Environment	Local Machine (Windows 11, Python 3.14, Flask)
Test Date	2 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Submit a valid loan application	1	10	Loan eligibility prediction displayed successfully
2	Multiple users submitting loan applications simultaneously	20	60	System handles multiple requests without failure
3	Predict loan eligibility with different applicant details	50	120	Accurate predictions generated with acceptable response time

4	Continuous prediction requests	100	180	Application remains stable with no crashes
---	--------------------------------	-----	-----	--

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 seconds	Pass	Fast prediction response
2	Response Time (Max)	< 5 seconds	3.4 seconds	Pass	Within acceptable limit
3	Throughput (Req/sec)	>20 Req/sec	28 Req/sec	Pass	Good throughput
4	Error Rate	< 1%	0%	Pass	No failed requests
5	CPU Utilization	< 80%	42%	Pass	Efficient CPU usage
6	Memory Utilization	< 80%	48%	Pass	Stable memory usage

Step 4: Observations & Analysis

Key Findings:

The Smart Lender application successfully processed loan eligibility requests with low response time and high accuracy. The XGBoost machine learning model generated predictions quickly while maintaining stable application performance. The system remained responsive even when multiple users accessed the application simultaneously.

Bottlenecks Identified:

- Slight increase in response time during high concurrent requests.
- Email notification introduces a small delay after prediction.
- Performance depends on local system resources during testing.

Optimization Steps Taken:

- Optimized the XGBoost model for faster prediction.
- Reduced unnecessary data processing during request handling.
- Improved Flask application routing.
- Used SQLite for efficient storage of user application details.
- Implemented environment variables (.env) for secure configuration.

Step 5: Screenshots / Evidence

Home Page:

Smart Lender

AI-Powered Loan Approval Prediction System Architecture

Evaluate commercial loan underwriting criteria instantly using optimized machine learning modeling layers.

Launch Evaluation Form

XGBoost Engine

Leverages high-performance gradient boosted decision trees for real-time risk assessment parameters and continuous feature importance auditing.

Automated Logging

Every parsed user submission profile dynamically tracks directly to transaction relational storage files for offline record keeping and compliance checks.

Instant Notification

Dispatches automated status updates and complete data profile summaries back to client communication layers instantly via custom server mailing channels.

Loan Eligibility Form:

Loan Application Assessment

Fill out the form fields completely to run the predictive qualification layer.

Full Name

John Doe

Email Address

name@example.com

Annual Income (₹)

e.g. 65000

Credit Score (300 - 850)

e.g. 720

Requested Loan Amount (₹)

e.g. 30000

Years Employed

e.g. 4

Risk Points (0-100)

e.g. 45.5

Evaluate Eligibility Profile

[← Return to Home Screen](#)

Prediction Result (Rejected):

SYSTEM ANALYSIS RESULT

Sorry divya, your Loan application has been Denied.

DECISION BREAKDOWN:

This assessment was reached primarily due to a low credit score profile.

An automated breakdown details summary and dynamic compliance confirmation log file copies have been dispatched via network transmission routing paths directly to the email address provided.

Evaluate Another Application

[Exit Dashboard Platform](#)

SYSTEM ANALYSIS RESULT

Congratulations sirisha! Your Loan is Approved.

DECISION BREAKDOWN:

This assessment was reached supported primarily by an excellent income-to-loan safety ratio.

An automated breakdown details summary and dynamic compliance confirmation log file copies have been dispatched via network transmission routing paths directly to the email address provided.

[Evaluate Another Application](#)

[Exit Dashboard Platform](#)